

## **Raw LIAR splits**

train.tsv / valid.tsv / test.tsv (12,836 rows)



## **Split-local cleaning**

whitespace and empty-string normalisation, dtype checks



## **Deduplication**

on id only; repeated statement text retained



## **Null handling**

drop core-null rows; impute metadata with 'Unknown'



## **Text statistics**

char length, word count, mean word length



## **Leak-free credit history**

train labels only, out-of-fold, prior-smoothed



## **Split-safe scaling**

scalers fitted on train, applied to valid and test



## **Feature union**

word TF-IDF + char TF-IDF + numeric + one-hot + subject



## **Classifier bank**

LR, SVM, SGD, MNB, RF, ET; six-class and binary



## **Protocol**

select on validation, score test once